



The Win Ratio and Beyond

Evolution, Applications, and Open Questions in Composite Endpoint Analysis

LU MAO

Imao@biostat.wisc.edu

Department of Biostatistics & Medical Informatics

University of Wisconsin-Madison

Outline

- Introduction
- Properties and Use Cases
- Extensions and Variations
- Challenges and Open Questions
- Conclusion

Introduction

Composite Endpoint

- **Traditional composite endpoint**

- **Time to first event**

- Major adverse cardiac event (MACE): death, heart failure, myocardio-infarction, stroke, etc.

- **Limitations:**

- Lack of clinical priority
 - Statistical inefficiency (waste of data)

- **Hierarchical composite endpoint (HCE)**

- **Example:** Death > nonfatal MACE > six-minute walk test (6MWT)/ NYHA class

Win Ratio: Basics

- **A common HCE**

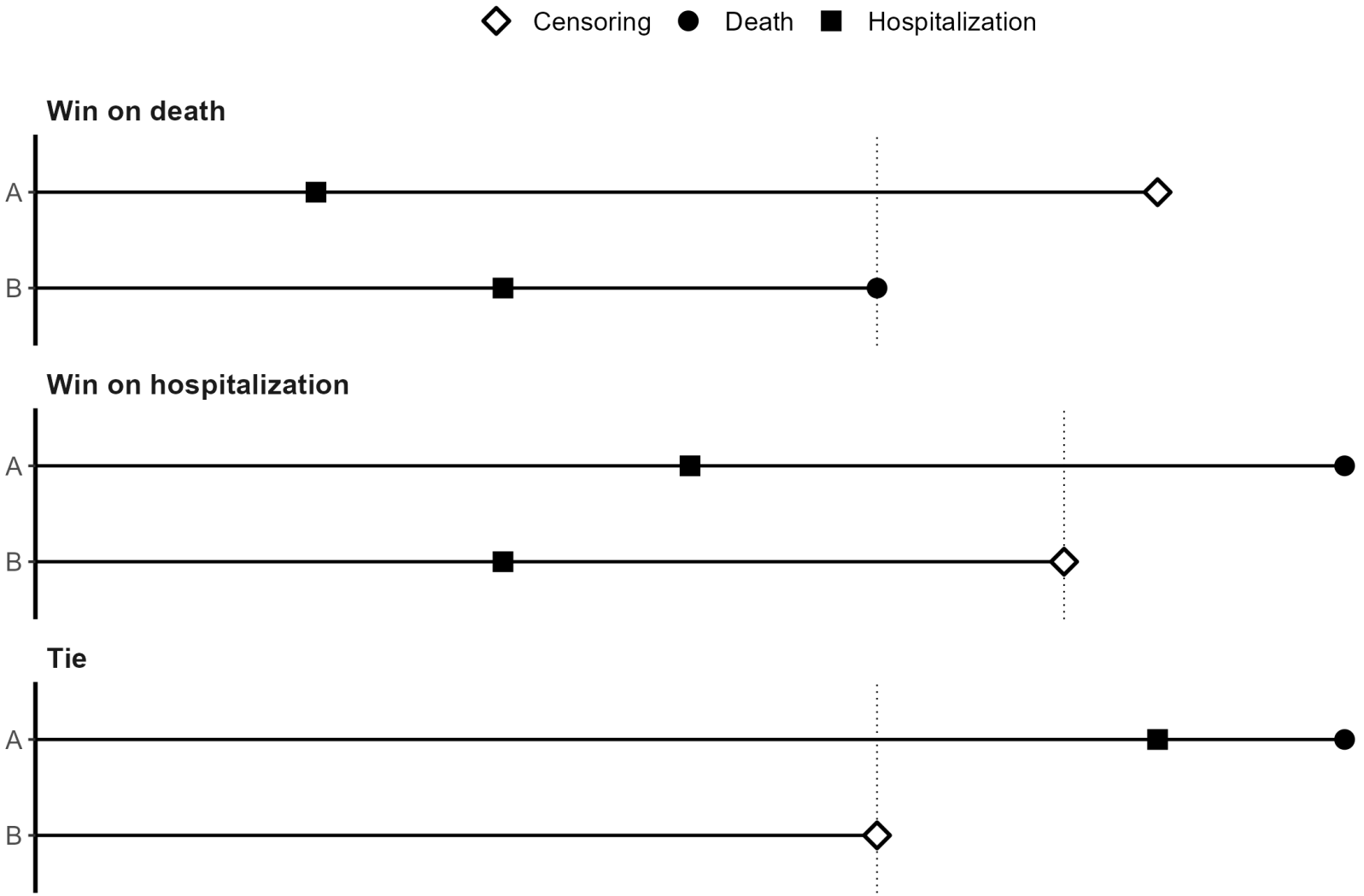
- **Proposed and popularized** by Pocock et al. ([2012](#))
- **Treatment vs control:** generalized pairwise comparisons (GPC) ([Buyse et al., 2025](#))
 - **Win-loss:** Longer survival → fewer/later nonfatal MACE → better 6MWT/NYHA
- **Effect size:** $WR = \text{wins} / \text{losses}$
 - Fold increase in probability of winning over losing

- **Alternative metrics**

- **Net benefit** (Proportion in favor): $NB = \text{wins} - \text{losses}$ ([Buyse, 2010](#))
- **Win odds:** $WO = (\text{wins} + 2^{-1}\text{ties}) / (\text{losses} + 2^{-1}\text{ties})$ ([Brunner et al., 2021](#))

Prioritized Comparison

Cut off at earlier end of follow-up



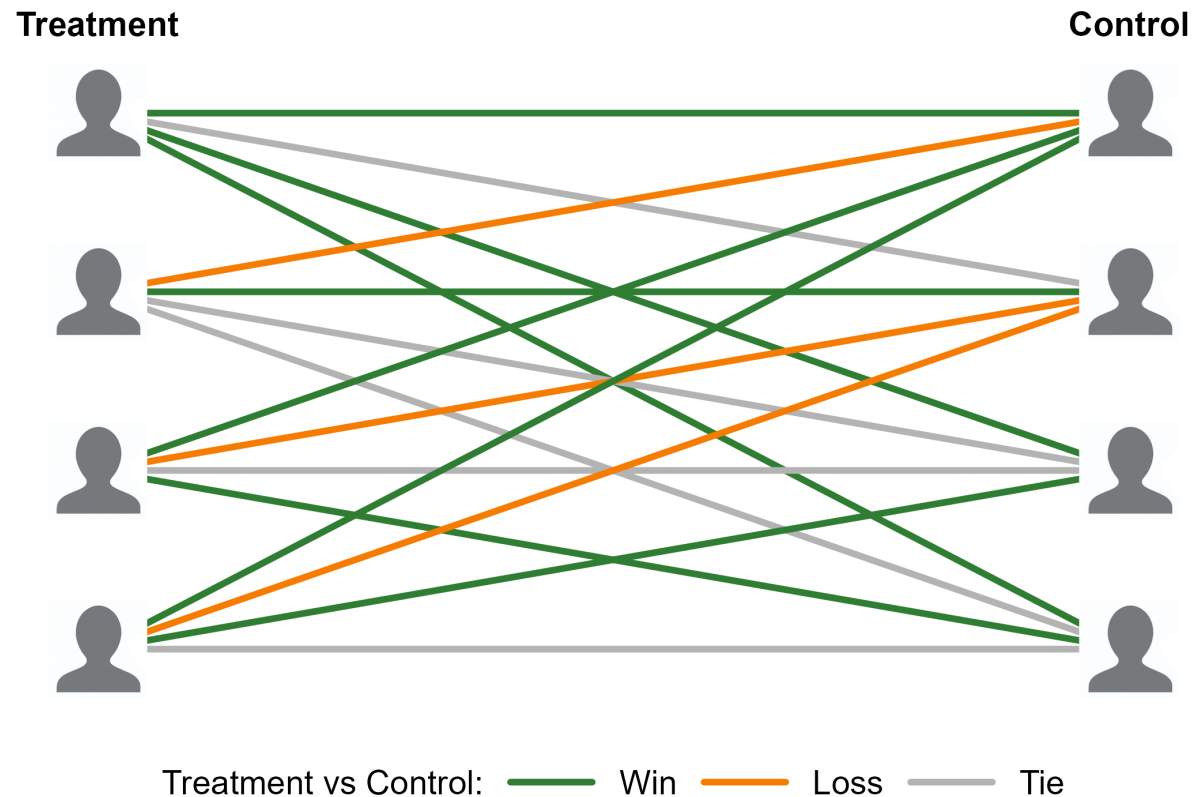
Pairwise Comparison

- **GPC Results**

- Wins: $8 / 16 = 50\%$
- Losses: $3 / 16 = 18.8\%$
- Ties: $5 / 16 = 31.2\%$

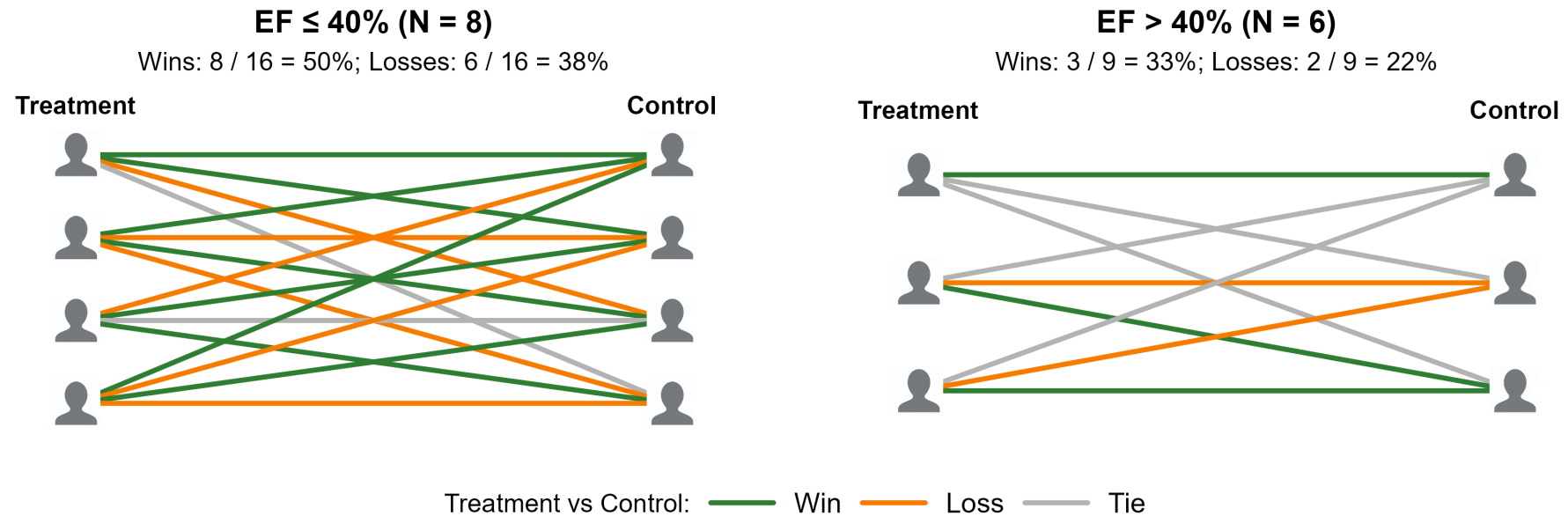
- **Effect Sizes**

- $WR = 50\% / 18.8\% = 2.67$
- $NB = 50\% - 18.8\% = 31.2\%$
- $WO = (50\% + 15.6\%) / (18.8\% + 15.6\%) = 1.72$



Stratification

- GPC within stratum → combine (Dong et al., 2018, 2023; Gasparyan et al., 2020)



$$\begin{aligned}\text{Wins: } & 50\% \times 8 / (8 + 6) + 33\% \times 6 / (8 + 6) = 42.7\% \\ \text{Losses: } & 38\% \times 8 / (8 + 6) + 22\% \times 6 / (8 + 6) = 31.1\%\end{aligned}$$

- Overall WR: $42.7\% / 31.1\% = 1.37$

Properties and Use Cases

General Setting

- Notation

- n_1, n_0 : sample sizes in treatment and control
- W_{ij} : win-loss indicator of subject i in treatment vs j in control

$$W_{ij} = \begin{cases} 1, & \text{if subject } i \text{ in treatment wins} \\ -1, & \text{if subject } j \text{ in control wins} \\ 0, & \text{if tie} \end{cases}$$

- Wins : $\hat{w}_{1,0} = (n_1 n_0)^{-1} \sum_{i=1}^{n_1} \sum_{j=1}^{n_0} I(W_{ij} = 1)$
- Losses : $\hat{w}_{0,1} = (n_1 n_0)^{-1} \sum_{i=1}^{n_1} \sum_{j=1}^{n_0} I(W_{ij} = -1)$

- $WR = \hat{w}_{1,0} / \hat{w}_{0,1}$; $NB = \hat{w}_{1,0} - \hat{w}_{0,1}$; $WO = (1 + NB) / (1 - NB)$

Binary Case Equivalencies

- **Binary outcome, no censoring**

- $Y_i^{(a)} = 1$ (response) vs 0 (no-response) for subject i in group a ($a = 0, 1$)
- Win-loss (1-0) fractions

$$\hat{w}_{a,1-a} = (n_a n_{1-a})^{-1} \sum_{i=1}^{n_a} \sum_{j=1}^{n_{1-a}} Y_i^{(a)} (1 - Y_j^{(1-a)}) = \hat{p}_a (1 - \hat{p}_{1-a})$$

- $\hat{p}_a = n_a^{-1} \sum_{i=1}^{n_a} Y_i^{(a)} = \text{response rate in group } a$

- **GPC stats**

- $WR = [\hat{p}_1(1 - \hat{p}_0)] / [\hat{p}_0(1 - \hat{p}_1)] = \text{response odds ratio}$
- $NB = \hat{p}_1 - \hat{p}_0 = \text{response rate difference}$

Variance Estimation

- **Bootstrap**

- Initial proposal ([Pocock et al., 2012](#))

- **Analytic asymptotic variance**

- Asymptotic decomposition of U -statistics ([Bebu & Lachin, 2016](#); [Dong et al., 2016](#); [Luo et al., 2015](#))
- Influence function-based
- **R packages:** [WR](#), [WINS](#), [BuyseTest](#), [WinRatio](#), etc.

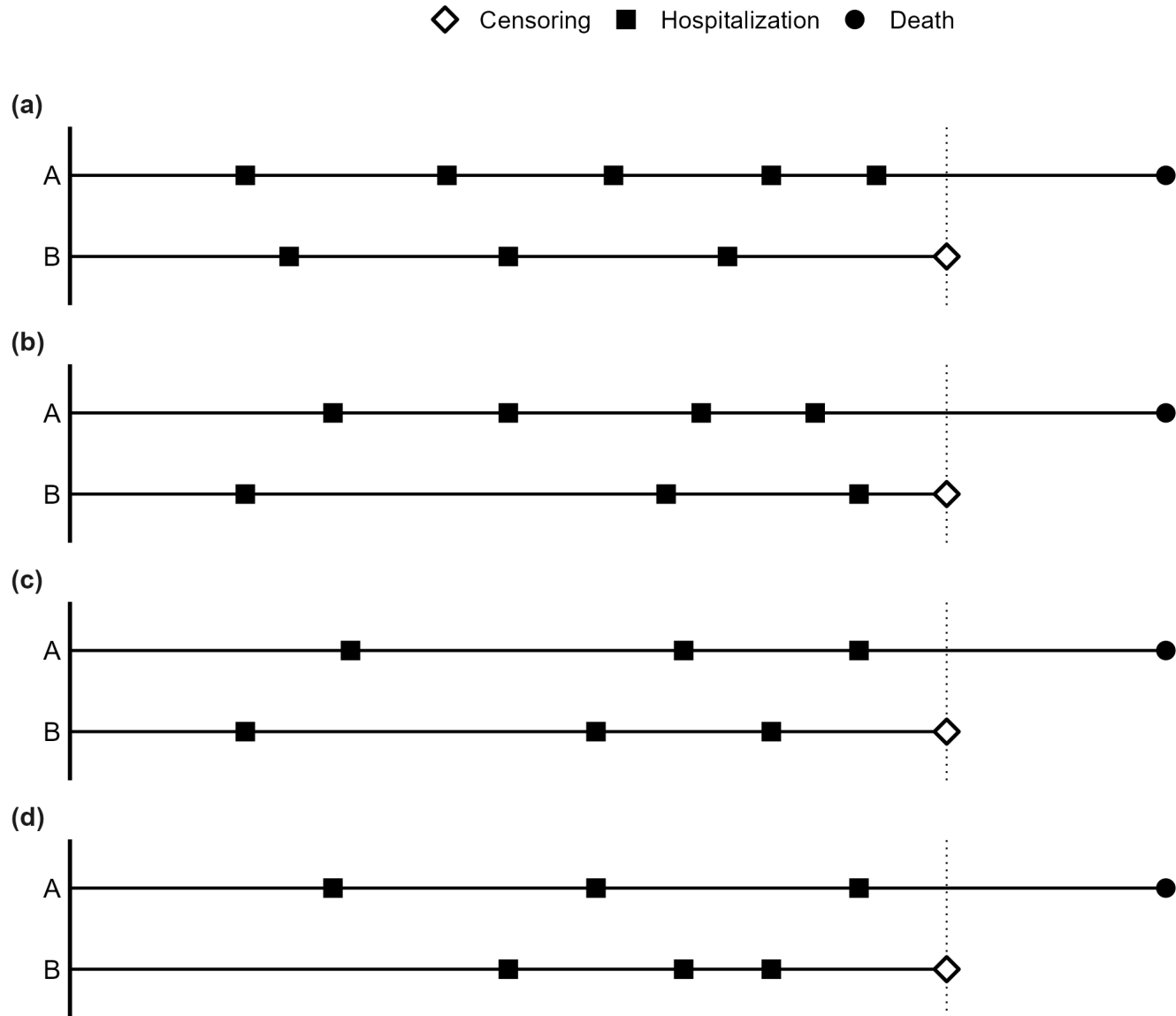
Null vs Alternative Hypotheses

- **Null hypothesis:** No treatment effect on any event
 - $WR = 1, NB = 0, WO = 1$ regardless of follow-up time
- **Alternative hypothesis**
 - Death and hospitalizations *stochastically later* in treatment than in control ([Luo et al., 2015](#); [Mao, 2019](#))
 - Stratified test: Stochastical order in each stratum
 - So $WR > 1, NB > 0, WO > 1$ regardless of follow-up time
 - Power $\rightarrow 1$ as $n = n_1 + n_0 \rightarrow \infty$

Recurrent Events (I)

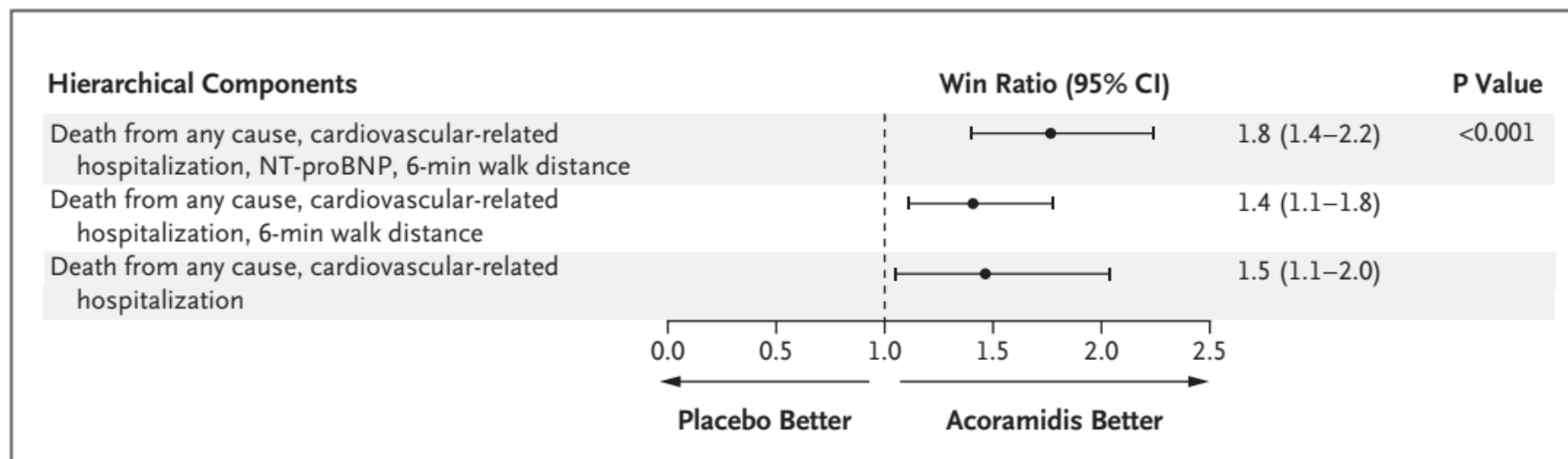
- **GPC to accommodate recurrent events**
 - E.g., hospitalizations, exacerbations, infections, etc.
- **Comparison rules**
 - Death $\rightarrow$ Number of events $\rightarrow$ Time to first/last event (Mao, Kim, & Li, 2022) $\rightarrow$... (Biomarker/QoL changes)
 - H_A : Treatment stochastically delays all events

Recurrent Events (II)



ATTRibute-CM (NEJM 2024)

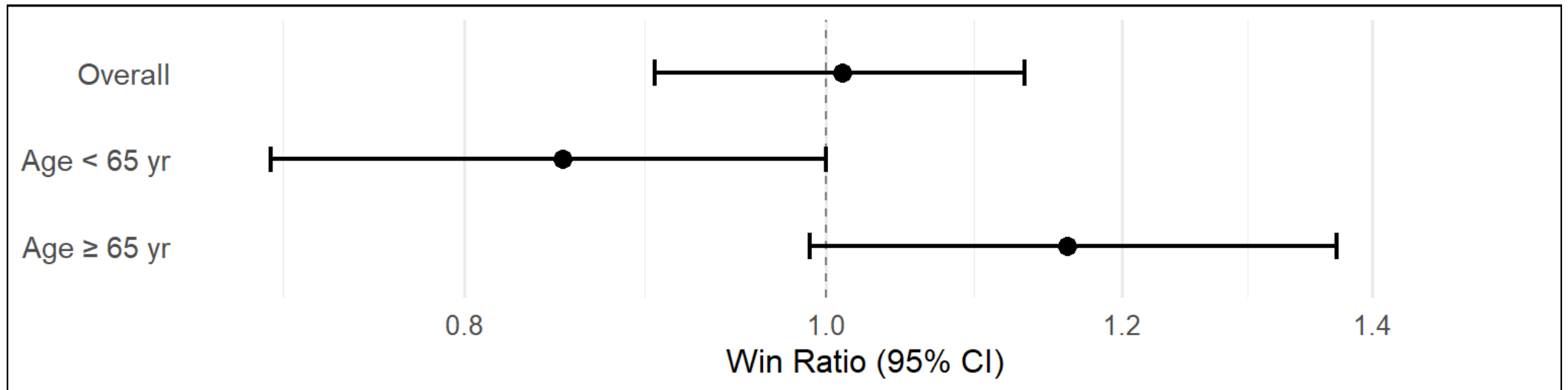
- Trial data ([Gillmore et al., 2024](#))
 - **Design:** Phase 3, double-blind RCT (n = 632; 2:1 randomization)
 - **Population:** Transthyretin amyloid cardiomyopathy (90 % wild-type, mean age 77 yr)
 - **Intervention:** Acoramidis 800 mg BID vs placebo for 30 months
 - **Primary HCE:** Death $\xrightarrow{71\%}$ CV hospitalization $\xrightarrow{42\%}$ NT-proBNP $\xrightarrow{12\%}$ 6MWT $\xrightarrow{0.3\%}$ ties
 - Stratified by TTR genotype, NT-proBNP level, eGFR, etc.



DEFENDER (JAMA 2024)

- Trial data ([Tavares et al., 2024](#))

- **Design:** Multicenter, open-label RCT (n = 507; 22 ICUs in Brazil)
- **Population:** Critically ill adults with an organ dysfunction (respiratory/ cardiovascular/ renal)
- **Intervention:** Dapagliflozin 10 mg daily vs standard care alone (up to 14d)
- **Primary HCE:** Hospital mortality $\xrightarrow{42\%}$ KRT initiation $\xrightarrow{35\%}$ ICU length of stay $\xrightarrow{4\%}$ ties



WR as Primary Endpoint (I)

Gregson et al. (2025) summarizes recent/ongoing RCTs:

Trial	Condition	Interventions	HCE
ATTRibute-CM (2024)	Transthyretin amyloid cardiomyopathy	Acoramidis vs placebo	Death → CV hosp → NT-proBNP/6MWT
COVID-PACT (2022)	Critically ill COVID-19	Full-dose AC vs standard-dose AC	Venous/arterial death → PE → DVT → ...
DAPA-MI (2024)	Clinically stable acute MI	Dapagliflozin vs placebo	CV death → non-CV death → HF hosp → ...
DEFENDER (2024)	Critical illness with organ failure	Dapagliflozin + SOC vs SOC	In-hospital death → KRT initiation → ICU LOS
EMPULSE (2022)	Acute HF (hospitalized)	Empagliflozin vs placebo	All-cause death → HF events (number/time) → KCCQ change
HEART-FID (2023)	HFrEF with iron deficiency	Ferric carboxymaltose vs placebo	All-cause death → HF hosp → 6MWT change

WR as Primary Endpoint (II)

Trial	Condition	Interventions	HCE
PARTNER 3 (2023)	Low-risk symptomatic AS	TAVR vs SAVR	Death → stroke → rehospitalization days
REDUCE LAP-HF II (2022)	HFpEF/HFmrEF	Atrial shunt device vs sham	CV death/stroke → HF hosp → KCCQ change
TRILUMINATE (2023)	Symptomatic tricuspid regurgitation	TEER vs OMT	Death/surgery → HF hosp → KCCQ change
VIP-ACS (2022)	Acute coronary syndrome	High-dose influenza vaccine vs standard	Death → MI → stroke → ...

...

Extensions and Variations

Sample Size Calculation

- Rank-based approximation (Yu & Ganju, 2022)

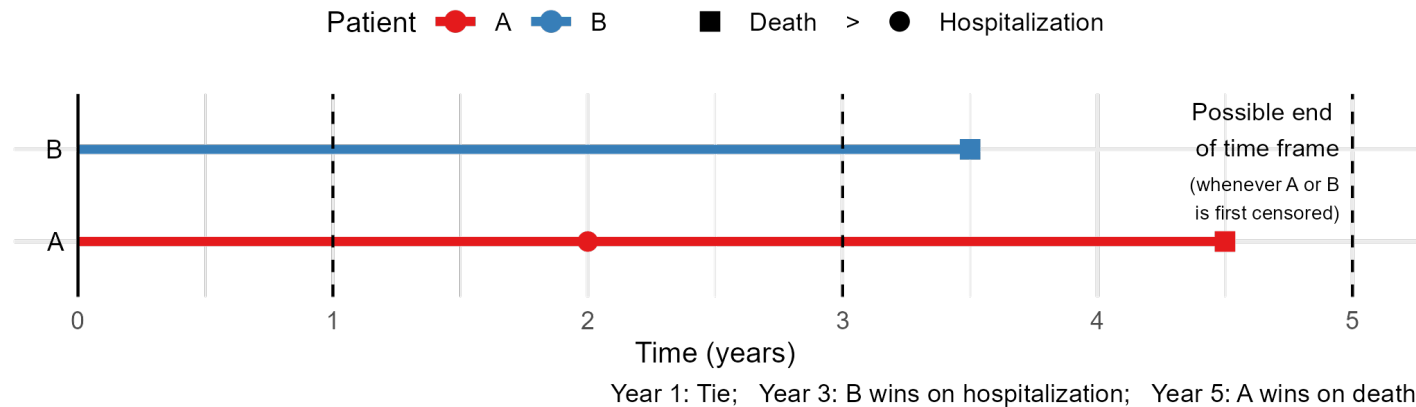
$$n \approx \frac{4(1 + p_{\text{tie}})(z_{1-\alpha/2} + z_{1-\beta})^2}{3q(1 - q)(1 - p_{\text{tie}}) \log^2(\text{WR})}$$

- $\alpha = 0.05$: type I error; $1 - \beta = 80\%, 90\%$: desired power
 - $q = n_1/n$; p_{tie} : proportion of ties
 - $\log(\text{WR})$ and p_{tie} elicitable from component-wise win-loss measures (Barnhart et al., 2025a, 2025b)
- Parametric approach (Mao, Kim, & Miao, 2022)
 - Gumbel–Hougaard copula + numerical integration (R package [WR](#))

Estimand Issue

- Without censoring

- WR, NB, WO = functions of joint distribution of events
- Change with follow-up time ([Mao, 2024](#); [Oakes, 2016](#))



- With censoring

- Weighted average of win-loss probabilities over time
- Trial-dependent estimand ([ICH, 2020](#))

Time Restriction

- **Specify time horizon τ**
 - Imagine all subjects followed up to τ (1 yr, 3 yr, etc.)
- **GPC under consistent time frame**
 - Estimand: $WR(\tau)$, $NB(\tau)$, $WO(\tau)$
 - Interpretation: win ratio/net benefit/win odds up to time τ
- **Estimation with censored data**
 - Inverse probability censoring weighting ([Dong et al., 2020b, 2021](#))
 - R-package: [WINS](#)
 - Multiple imputations ([T. Wang et al., 2023, 2024](#))

Regression Analysis

- A proportional win-fractions (PW) model

$$\frac{\text{pr}(W_{ij}(t) = 1 \mid Z_i, Z_j)}{\text{pr}(W_{ij}(t) = -1 \mid Z_i, Z_j)} = \exp\{\beta^T (Z_i - Z_j)\}$$

- $W_{ij}(t)$: win-loss indicator by time t
- Z_i, Z_j : covariate vectors for subjects i and j
- β : log-WRs for one-unit increases in covariates
- U-statistic estimation; residual analysis to check WR constancy ([Mao & Wang, 2021](#), R package [WR](#)); stratification ([T. Wang & Mao, 2022](#))
- Elastic net regularization for high-dimensional Z ([Mao, 2025](#), R tool [WRNet](#))

Win-Loss Times

- Restricted mean time in favor of treatment

$$\mu(\tau) = w_{1,0}(\tau) - w_{0,1}(\tau)$$

- $w_{a,1-a}(\tau)$: average *time* group a wins over group $1 - a$ up to time τ
- **Interpretation:** net favorable time gained/lost by treatment ([Mao, 2023](#), R package [rmt](#))
 - Decomposable by component
- **Other win time measures**
 - Expected win time ($\tau \rightarrow$ maximum length of follow-up), etc. ([Troendle et al., 2024](#))

DOOR

- **Desirability of outcome ranking** (Evans et al., 2015)

- Pre-defined ordinal categories of clinical outcome $Y^{(a)}$
 1. Dead
 2. Alive with severe AE
 3. Alive with mild AE
 4. Alive without AE

- **Estimand**

- Tie-adjusted *win probability* from GPC ranked by DOOR

$$\theta_{\text{DOOR}} = \Pr(Y^{(1)} > Y^{(0)}) + \frac{1}{2} \Pr(Y^{(1)} = Y^{(0)})$$

- R package [door](#)

Challenges and Open Questions

Critiques

- **Pitfalls and caveats** (Butler et al., 2024)
 - WR ignores ties
 - Win relative to *loss* rather than “*not win*”
 - Some consider this to *overstate* effect size
 - Inclusion of low-priority biomarkers
 - Surrogate measures drive treatment effect
 - Subjectivity in defining HCE
 - Clinical input needed
 - Post hoc construction inflates type I error

Covariate Adjustment

- **Different from regression**

- **Marginal estimands** between groups, not conditioning on Z
- **Gain efficiency** when outcome-covariate model is true, otherwise still valid (robustness)
- **Standard endpoints**
 - Continuous, binary, univariate survival, etc. (Tsiatis et al., 2008; B. Wang et al., 2021; Ye et al., 2023)
 - FDA recommendation (FDA, 2023)

- **Challenges with WR**

- U -statistic structure
- Lack of likelihood structure

Interim Analysis

- **Purpose**

- Analyze interim data for evidence of efficacy/futility → stop trial early
- Univariate survival: information accrued $\propto$ number of events

- **Challenges with WR**

- Information time not event-driven
 - Case study: RMST ([Lu & Tian, 2021](#); [Luo et al., 2019](#))
- Correlations between component events
 - *Dependent* increments

Meta Analysis

- Challenges

- Primary studies *not* reporting win-loss measures
- Primary studies over *different* time spans
- Primary studies with *different* definitions of win/loss

- WinKM: A toolkit to start

- Calculate win-loss statistics based on
 - KM estimates for OS and EFS
 - At-risk table at selected time points
 - Total event counts (reported in the CONSORT diagram or results section)

Conclusion

Key Takeaways (I)

- **Win ratio and its variants**

- Intuitive, clinically interpretable effect sizes for HCEs
- Support recurrent events, stratification, etc.
- Time-restricted estimands, regression analysis, win times, DOOR, etc.

- **Good practice**

- Pre-specify HCE definition and analysis plan
- Caution in including surrogate measures
 - Anticipated clinical event rates
 - Set thresholds for meaningful differences
- Sensitivity analyses

Key Takeaways (II)

- R packages

- `WR`, `WINS`, `BuyseTest`, `WinRatio`, `rmt`, `WRNet`, `door`, `WinKM`, etc.

- Open questions

- Covariate adjustment, interim analysis, meta analysis, etc.

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